THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The point $(5, -2)$ is reflected in the y-axis. The image is: 1

(a) $(-5, -2)$ **(b)** $(-5, 2)$ **(c)** $(-2, -5)$ **(d)** $(5, -2)$

2. Which of these is a linear polynomial? 1

(a) $3x + 4$ **(b)** $x^3 - x$ **(c)** 7 **(d)** $x^2 + 4$

3. The number of decimal places in the expansion of $\frac{3}{2^3 \times 5^2}$ is: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

4. $(2x + 3y)^2 =$ 1

(a) $4x^2 - 12xy + 9y^2$ **(b)** $4x^2 + 12xy + 9y^2$ **(c)** $4x^2 + 9y^2$ **(d)** $4x^2 + 6xy + 9y^2$

5. A chord is at distance 7 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 47 **(b)** 48 **(c)** 49 **(d)** 50

6. The area of a triangle with sides 13 cm, 14 cm and 15 cm is (cm^2): 1

(a) 86 **(b)** 85 **(c)** 84 **(d)** 83

7. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 2 is: 1

(a) $\frac{39}{50}$ **(b)** $\frac{11}{50}$ **(c)** $\frac{11}{51}$ **(d)** $\frac{6}{25}$

8. If $t_n = 2n^2 + 3$, then t_5 is: 1

(a) 55 **(b)** 53 **(c)** 52 **(d)** 54

9. The converse of the proposition "If a number is divisible by 6, then it is divisible by 3" is: 1

(a) If it is not true that a number is divisible by 6, then it is not true that it is divisible by 3.
(b) If a number is divisible by 3, then it is divisible by 6.
(c) If a number is divisible by 6, then it is divisible by 3.
(d) If it is not true that it is divisible by 3, then a number is divisible by 6.

10. The average marks of 27 students is 57 and of another 39 students is 35. The average of all 66 students is: 1

(a) 44 **(b)** 45 **(c)** 43 **(d)** 46

11. Adding 1747 and 4333 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 4 **(b)** 1 **(c)** 3 **(d)** 2

12. The angles of a quadrilateral are in the ratio $1:2:6:6$. The largest angle is: 1

(a) 134° **(b)** 164° **(c)** 154° **(d)** 144°

13. Is the point $(-3, 2)$ a solution of $x + 2y = 2$? 1

(a) No **(b)** Cannot be said **(c)** Only for $x = 0$ **(d)** Yes

14. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 3.5 cm and height 11 cm is: 1

(a) 292 cm^2 **(b)** 242 cm^2 **(c)** 267 cm^2 **(d)** 217 cm^2

15. The distance between $(1, 2)$ and $(4, 6)$ is: 1

(a) 6 **(b)** 4 **(c)** 7 **(d)** 5

16. If $x = 5$ is a zero of the polynomial $2x + k$, then $k =$ 1

(a) -10 **(b)** -11 **(c)** -9 **(d)** -8

17. $x = 0.\overline{27}$ equals: 1

(a) $\frac{3}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{11}$

18. $(3x - 2y)^2 =$ 1

(a) $9x^2 + 4y^2$ **(b)** $9x^2 + 12xy + 4y^2$ **(c)** $9x^2 - 12xy + 4y^2$ **(d)** $9x^2 - 6xy + 4y^2$

19. The radius of a circle is 25 cm and a chord is 40 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 15 **(b)** 14 **(c)** 17 **(d)** 16

20. The semi-perimeter of a triangle with sides 13 cm, 14 cm and 15 cm is (cm): 1

(a) 21 **(b)** 23 **(c)** 20 **(d)** 22

21. In a survey of 500 families, 140 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{7}{25}$ **(b)** $\frac{7}{26}$ **(c)** $\frac{8}{25}$ **(d)** $\frac{18}{25}$

22. Which term of the sequence $t_n = 3n + 2$ is 53? 1

(a) 17th term **(b)** 16th term **(c)** 18th term **(d)** 19th term

23. The proposition “If a number is divisible by 6, then it is divisible by 3” is true. Its converse “If it is divisible by 3, then a number is divisible by 6” is: 1

(a) True only for some values **(b)** It cannot be decided
(c) True **(d)** False

24. The average of 14 numbers is 31. When a number 61 is included, the average of all 15 numbers becomes: 1

(a) 33 **(b)** 35 **(c)** 34 **(d)** 32

25. The divisor algorithm applied to 44 produces a list with how many entries? 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 5

26. In a parallelogram, two opposite angles are $(2x + 11)^\circ$ and $(x + 18)^\circ$. The value of x is: 1

(a) 7 **(b)** 8 **(c)** 9 **(d)** 6

27. The y-intercept of the line $x + 6y = 24$ is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

28. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 3.5 cm and height 27 cm is: 1

(a) 807 cm^2 **(b)** 739 cm^2 **(c)** 671 cm^2 **(d)** 603 cm^2

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29. The midpoint of the segment joining $(-4, 6)$ and $(8, -2)$ is: 1

(a) $(-2, 2)$ **(b)** $(-2, -2)$ **(c)** $(2, -2)$ **(d)** $(2, 2)$

30. The value of $3x + 5$ at $x = -2$ is: 1

(a) 0 **(b)** -1 **(c)** 1 **(d)** -2

31. An irrational number between 2 and 3 is: 1

(a) $\sqrt{11}$ **(b)** $\sqrt{3}$ **(c)** $\sqrt{5}$ **(d)** $\sqrt{10}$

32. $(x + 7)(x - 3) =$ 1

(a) $x^2 + 10x - 21$ **(b)** $x^2 + 4x + 21$ **(c)** $x^2 + 4x - 21$ **(d)** $x^2 - 4x - 21$

33. The chord AB subtends 74° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 37° **(b)** 47° **(c)** 57° **(d)** 27°

34. The area of a circle with circumference 88 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 617 **(b)** 616 **(c)** 615 **(d)** 618

35. A letter is chosen at random from the word EDUCATION. The probability that it is a vowel is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{5}{9}$

36. A sequence has $t_1 = 4$ and $t_{n+1} = t_n + 3$. Then t_6 equals: 1

(a) 19 **(b)** 20 **(c)** 18 **(d)** 21

37. Which of the following is a counterexample to the statement "If $a < b$, then $a^2 < b^2$"? 1

(a) $a = 3,\ b = 6$ **(b)** $a = 1,\ b = 4$ **(c)** $a = -3,\ b = 2$ **(d)** $a = 2,\ b = 5$

38. Marks of 70, 65 and 75 are given weights 6, 6 and 3 respectively. The weighted average is: 1

(a) 70 **(b)** 69 **(c)** 71 **(d)** 68

39. By Euclid's algorithm, gcd$(90, 82)$ is: 1

(a) 4 **(b)** 3 **(c)** 1 **(d)** 2

40. Two adjacent angles of a parallelogram are in the ratio $1 : 8$. The smaller angle is: 1

(a) 20° **(b)** 30° **(c)** 40° **(d)** 10°

41. The slope of the line through $(4, 7)$ and $(5, -2)$ is: 1

(a) -8 **(b)** -9 **(c)** -7 **(d)** -10

42. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 14 cm and slant height 50 cm is: 1

(a) 1979 cm^2 **(b)** 2642 cm^2 **(c)** 2200 cm^2 **(d)** 2421 cm^2

43. A point on the x-axis at distance 6 units to the right of the origin has coordinates: 1

(a) $(-6, 0)$ **(b)** $(6, 0)$ **(c)** $(0, 6)$ **(d)** $(7, 0)$

44. The line $y = 3x + 1$ passes through $(2, p)$. The value of p is: 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 9

45. The sum of a rational number and an irrational number is: 1

(a) zero **(b)** rational **(c)** irrational **(d)** an integer

46. By identity, $103^2 =$ 1

(a) 10609 **(b)** 10610 **(c)** 10608 **(d)** 10611

47. The chord AB subtends 66° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 147° **(b)** 137° **(c)** 167° **(d)** 157°

48. A wheel of radius 28 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 177 **(b)** 175 **(c)** 176 **(d)** 178

49. Three coins are tossed together. The probability of getting exactly one head is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{8}$

50. The 15th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 46 **(b)** 49 **(c)** 47 **(d)** 48

END OF QUESTION PAPER **Total: 50 Marks**

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